Math 126 **Opportunity B** Name: ____________________

USC ID: ____________________

Instructions:

- Make sure to show each step of your solution for full credit - correct answers without work will not receive credit. Some questions may remind you to show your work, but you should assume each question requires you to justify your steps by making your thought process clear in your solution. You can use the solutions we write in class as an indication of how much work you are expected to show. You can ask the instructor if it is not clear how much work you are expected to show on a particular problem.
- Write all of your work and answers on this document. If you need extra paper, please ask the instructor.
- Indicate your affirmation of the following statement by signing below.

I did not give or receive any unauthorized assistance on this exam. I understand that the penalty for academic dishonesty is an F for the course and possible suspension or expulsion from the university.

Signature: ____

Moments and centroid

$$M_x = \frac{\rho}{2}\int_a^b (f(x)^2 - g(x)^2)dx$$
$$M_y = \rho\int_a^b x(f(x) - g(x))dx$$
$$m = \rho\int_a^b (f(x) - g(x))dx$$
$$(\overline{x}, \overline{y}) = \left(\frac{M_y}{m}, \frac{M_x}{m}\right)$$

Area between curves

$$A = \int_a^b (f(x) - g(x))dx$$

Work

$$W = \int_a^b F(x)dx$$

Hydrostatic force

$$F = \rho g\int_a^b w(y)d(y)dy$$

Surface area

$$A = \int_a^b 2\pi f(x)\sqrt{1 + f'(x)^2}dx$$
$$A = \int_a^b 2\pi x\sqrt{1 + f'(x)^2}dx$$

Arc length

$$L = \int_a^b \sqrt{1 + f'(x)^2}dx$$

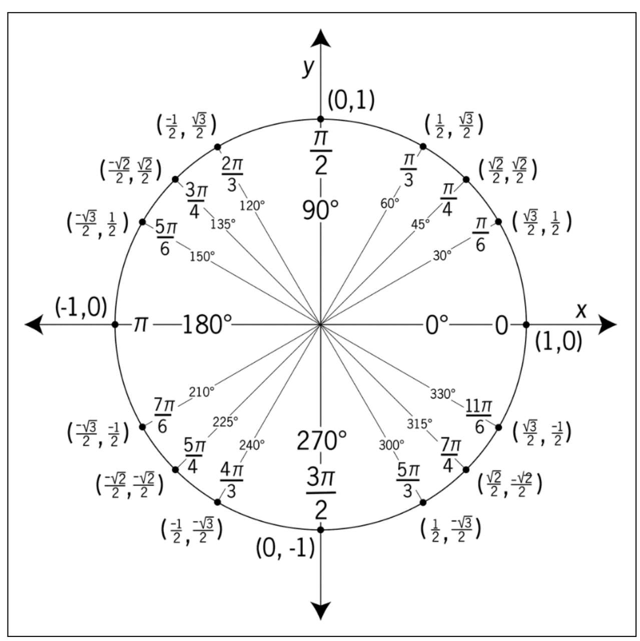
Volumes

Cross sections: $V = \int_a^b A(x)dx$

Cylindrical shells: $V = \int_a^b 2\pi x f(x)dx$

Polar coordinates

$$\frac{dy}{dx} = \frac{\frac{dr}{d\theta}\sin\theta + r\cos\theta}{\frac{dr}{d\theta}\cos\theta - r\sin\theta}$$
$$A = \int_a^b \frac{1}{2}(f(\theta)^2 - g(\theta)^2)d\theta$$
$$L = \int_a^b \sqrt{r^2 + \left(\frac{dr}{d\theta}\right)^2}d\theta$$

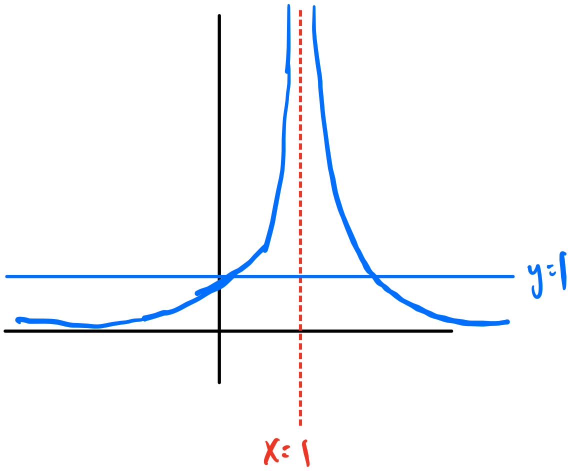


B1 - Areas between curves (inspired by past final worksheet problem 1 and Brightspace problem 5)

(a) Consider the region in the plane that satisfies

$$1 \leq y \leq \frac{1}{(x-1)^2}.$$

Write down two different integrals that represent the area of this region: one that integrates along the x-axis and one that integrates along the y-axis. You do not need to evaluate either integral.



$1 = \frac{1}{(x-1)^2} \rightarrow (x-1)^2 = 1 \rightarrow x-1 = \pm 1$

$\rightarrow x = 0, 2$

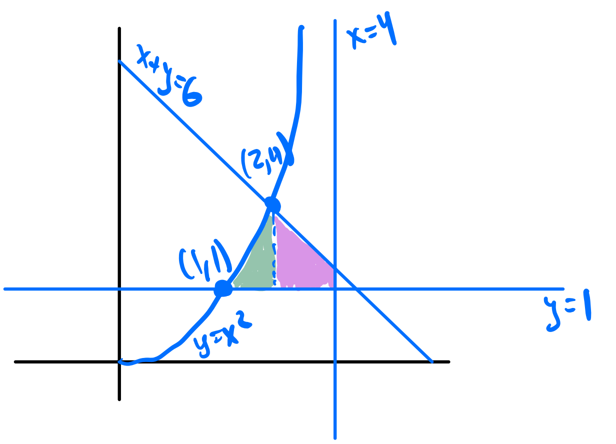
$$\boxed{x\text{-axis: } \int_0^2 \left(\frac{1}{(x-1)^2} - 1\right) dx}$$

$y = \frac{1}{(x-1)^2} \rightarrow (x-1)^2 = \frac{1}{y} \rightarrow x-1 = \pm\frac{1}{\sqrt{y}}$

$x = 1 \pm \frac{1}{\sqrt{y}}$

$$\boxed{y\text{-axis: } \int_1^\infty \left(1 + \frac{1}{\sqrt{y}} - \left(1 - \frac{1}{\sqrt{y}}\right)\right) dy}$$

(b) Find the area of the region bounded by $y = x^2$, $x + y = 6$, $x = 4$ and $y = 1$. Express your answer as an integral or sum of integrals. You do not need to evaluate any integral(s) you use.

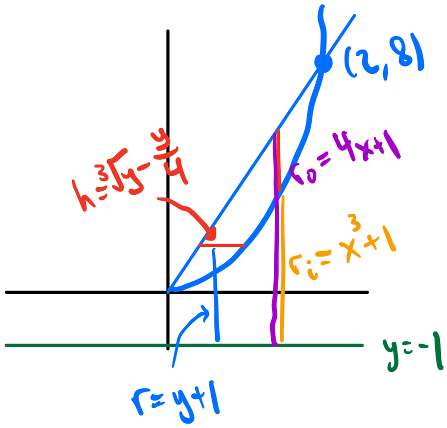


$\left.\begin{array}{l} y = 6 - x \\ y = x^2 \end{array}\right\}$ intersect at $x = 2$

$$\int_1^2 (x^2 - 1)\,dx + \int_2^4 (6 - x - 1)\,dx$$

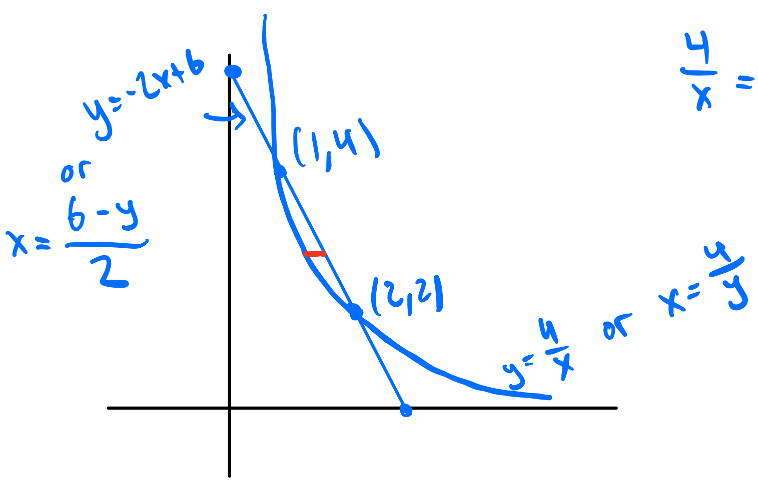
B2 - Volumes (inspired by part 3 of the class notes and Brightspace problem 3)

(a) The region bounded by $y = x^3$ and $y = 4x$ (with $x \geq 0$) is rotated about the line $y = -1$. Write two different integrals that represent the volume of the resulting solid: one that integrates along the x-axis and one that integrates along the y-axis. You do not need to evaluate either integral.



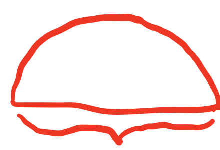
$$\boxed{\begin{array}{l} x\text{-axis: } \int_0^2 \left(\pi(4x+1)^2 - \pi(x^3+1)^2\right)dx \\ y\text{-axis: } \int_0^8 2\pi(y+1)\cdot\left(\sqrt[3]{y} - \frac{y}{4}\right)dy \end{array}}$$

(b) The base of a solid is the region bounded by $y = \dfrac{4}{x}$ and $y = -2x + 6$, and cross sections of the solid perpendicular to the y-axis are semicircles. Write an integral representing the volume of this solid. You do not need to evaluate the integral.



$$\frac{4}{x} = -2x+6 \rightarrow 2x^2 - 6x + 4 = 0$$

$$2(x-2)(x-1) = 0$$



$$\text{diameter} = \left(\frac{4}{y} - \frac{6-y}{2}\right)$$

$$\text{Area} = \frac{\pi}{2}\left(\frac{2}{y} - \frac{6-y}{4}\right)^2$$

$$\boxed{V = \int_2^4 \frac{\pi}{2}\left(\frac{2}{y} - \frac{6-y}{4}\right)^2 dy}$$

B3 - Arclength and surface area (inspired by Brightspace problem 10 and part 1 of the class notes)

(a) Suppose the arc length of a curve $f(x)$ from $x = 0$ to $x = 3$ is 5. Which of the following could be the arc length function of $f(x)$ starting at $x = 0$? Circle all that apply.

A. $s(x) = \frac{5x}{3}$

B. $s(x) = x + 2$

C. $s(x) = x^2 - \frac{5}{3}x$

D. $s(x) = x^2 - \frac{4}{3}x$

E. $s(x) = \frac{4}{3}x$

(b) Consider the curve $y = \frac{x^2}{4} - \frac{\ln x}{2}$ on the interval $2 \leq x \leq 5$.

(i) Find the arc length of this curve.

$y' = \frac{x}{2} - \frac{1}{2x}$ $\quad (y')^2 + 1 = \frac{x^2}{4} - \frac{1}{2} + \frac{1}{4x^2} + 1 = \left(\frac{x}{2} + \frac{1}{2x}\right)^2$

$\int_2^5 \sqrt{\left(\frac{x}{2} + \frac{1}{2x}\right)^2}\, dx = \int_2^5 \left(\frac{x}{2} + \frac{1}{2x}\right) dx = \frac{x^2}{4} + \frac{1}{2}\ln x \Big|_2^5 = \boxed{\frac{25}{4} + \frac{1}{2}\ln 5 - \left(1 + \frac{\ln(2)}{2}\right)}$

(ii) Find the exact area of the surface obtained by rotating this curve about the y-axis. You should evaluate any integral(s) you use, but you do not need to simplify your final answer.

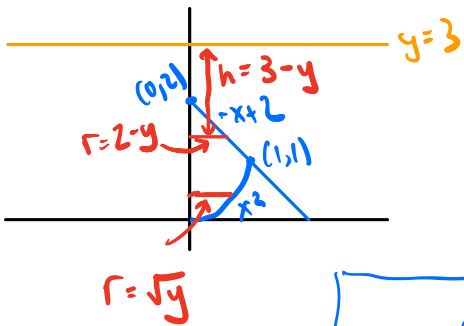
$S = \int_2^5 2\pi x \cdot \left(\frac{x}{2} + \frac{1}{2x}\right) dx$

$= 2\pi \int_2^5 \left(\frac{x^2}{2} + \frac{1}{2}\right) dx = 2\pi \left(\frac{x^3}{6} + \frac{1}{2}x\right)\Big|_2^5$

$\boxed{= 2\pi \left(\frac{5^3}{6} + \frac{1}{2}\cdot 5 - \left(\frac{2^3}{6} + \frac{1}{2}\cdot 2\right)\right)}$

B4 - Applications of integration to physics and engineering (inspired by past final worksheet problem 2 and Brightspace problem 9)

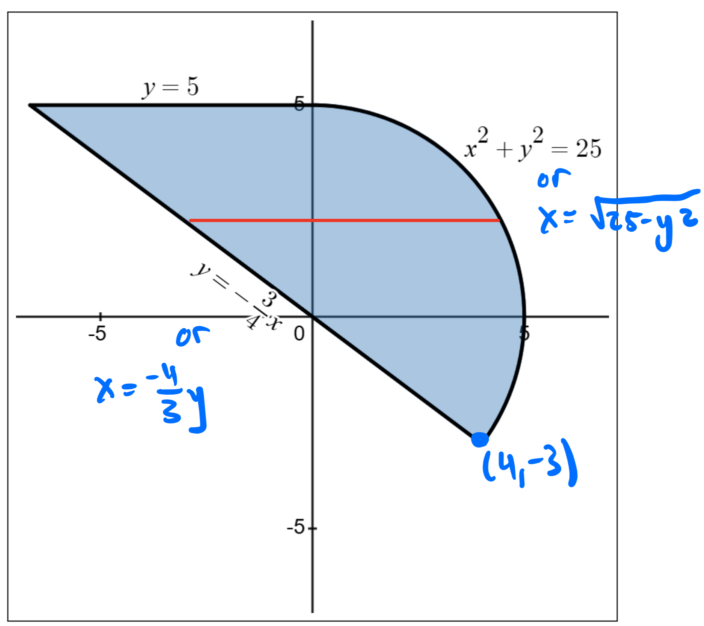
(a) Suppose that $y = 3$ represents ground level, and a tank buried under ground is in the shape of the region bounded by $y = -x + 2$, $x \geq 0$, and $y = x^2$ rotated about the y-axis. If this tank is filled with a liquid of density ρ, find the work needed to pump all of the liquid up to ground level. *Hint:* You will need to use two different integrals. You do not need to evaluate either integral. Use g for the acceleration of gravity.



$$dW = \rho g \, dV \cdot h = \rho g \pi r^2 dy \cdot h$$

$$\boxed{W = \int dW = \int_0^1 \rho g \pi (\sqrt{y})^2 (3-y)\, dy + \int_1^2 \rho g \pi (2-y)^2 (3-y)\, dy}$$

(b) The lamina on the right is defined by three equations. The equations for the straight lines are $y = 5$ and $y = -\frac{3}{4}x$, and the equation of the curved side is $x^2 + y^2 = 25$. Write integrals representing the moments of the lamina about the x and y axes. You do not need to evaluate the integrals.



$$x^2 + \left(-\frac{3}{4}x\right)^2 = 25$$

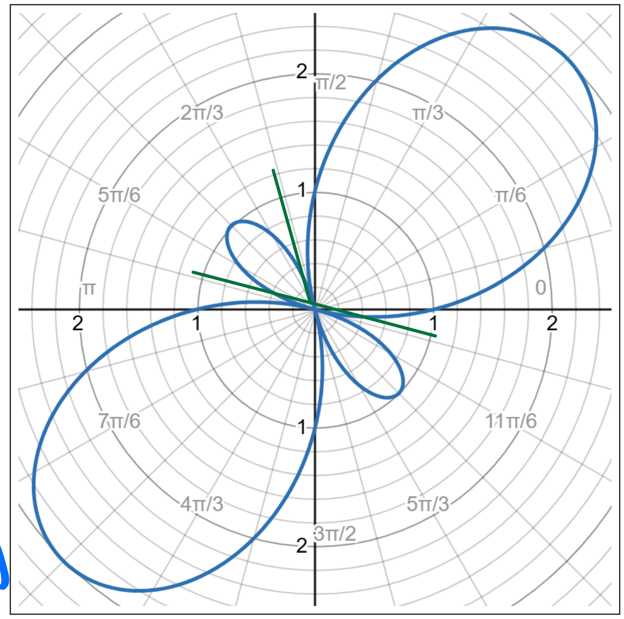
$$x^2 + \frac{9x^2}{16} = 25$$

$$\frac{25x^2}{16} = 25 \rightarrow x = 4$$
$$\rightarrow y = -3$$

$$\boxed{\begin{aligned} M_x &= \int_{-3}^{5} y \cdot \left(\sqrt{25-y^2} - \frac{-4}{3}y\right) dy \\ M_y &= \int_{-3}^{5} \frac{1}{2}\left(\left(\sqrt{25-y^2}\right)^2 - \left(\frac{-4}{3}y\right)^2\right) dy \end{aligned}}$$

B5 - Areas and lengths in polar coordinates (inspired by past final worksheet problem 1 and Brightspace problems 4 and 10)

The graph of the polar curve $r = 1 + 2\sin(2\theta)$ is shown to the right. It has two large petals and two small petals.



(a) Find the slope of the curve at $\theta = \frac{\pi}{2}$.

$$\frac{dy}{dx} = \frac{\frac{dr}{d\theta}\sin\theta + r\cos\theta}{\frac{dr}{d\theta}\cos\theta - r\sin\theta}$$

$$\frac{dr}{d\theta} = 4\cos(2\theta) \quad @ \ \theta = \frac{\pi}{2} \rightarrow -4$$

$$\frac{dy}{dx} = \frac{-4\sin(\frac{\pi}{2}) + (1+2\sin(2\cdot\frac{\pi}{2}))\cos(\frac{\pi}{2})}{-4\cos(\frac{\pi}{2}) - (1+2\sin(2\cdot\frac{\pi}{2}))\sin(\frac{\pi}{2})}$$

$$= \frac{-4}{-1} = \boxed{4}$$

(b) Find an integral that represents the area of one of the large petals. You do not need to evaluate the integral.

$1 + 2\sin(2\theta) = 0 \rightarrow \sin(2\theta) = \frac{-1}{2}$

for large petal, we want a solution in Q IV and Q II

$\rightarrow \quad 2\theta = \frac{-\pi}{6} \quad$ and $\quad 2\theta = \frac{7\pi}{6}$

$\theta = \frac{-\pi}{12} \qquad \theta = \frac{7\pi}{12}$

$$\boxed{A = \int_{-\frac{\pi}{12}}^{\frac{7\pi}{12}} \frac{1}{2}(1+2\sin(2\theta))^2\, d\theta}$$

(c) Find an integral that represents the arc length of one of the small petals. You do not need to evaluate the integral.

$\sin(2\theta) = \frac{-1}{2} \rightarrow$ for small petal we want two solutions in QII

$2\theta = \frac{7\pi}{6} \qquad 2\theta = \frac{11\pi}{6}$

$\theta = \frac{7\pi}{12} \qquad \theta = \frac{11\pi}{12}$

$$\boxed{L = \int_{\frac{7\pi}{12}}^{\frac{11\pi}{12}} \sqrt{(1+2\sin(2\theta))^2 + (4\cos(2\theta))^2}\, d\theta}$$